

ISTA 322 Final Project Part 1

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What is your goal?

What questions/types of questions will you be able to answer?:

My relational database will be able to answer music consumer based questions like:

- ✓ "What are the most popular albums in the last year in a specific genre?"
- ✓ "Can I find artists who are popular but haven't 'broken through' to the mainstream yet?"
- ✓ "What are the key characteristics (genre tags, popularity, etc.) of my favorite artist's music?"
- ✓ "What are some new albums that are trending among people who like a specific artist?"

What can your RDB be used for, who would the end user be?:

My relational database can be used as a backend for a music discovery platforms, specifically targeting the average music fan. These users want to discover new music that they will enjoy, and my database will connect an artist's basic info, popularity, and genres all into one source.

Data game plan:

List of data source you will use (links to them):

- ✓ *MusicBrainz API*: https://musicbrainz.org/doc/Development/XML_Web_Service/Version_2
- ✓ *Last.fm API*: <https://www.last.fm/api>
- ✓ *Spotify Web API*: <https://developer.spotify.com/documentation/web-api>

What data are available at each source:

From the *MusicBrainz API*, I will pull artist information (id, name, album id, etc.) and use this as a foundation for my schema. From the *Last.fm API* I will extract data like listener/play count and fan-generated tags which is helpful for details in niche genres. Then, from the *Spotify Web API* I will use the popularity score as well as related artists, using Spotify's own music discovery data to my advantage.

Data game plan (cont.):

What data will you use and how will you integrate it?:

I will begin by extracting a group of artists from one of my APIs as my core records and then try to replicate similar groups from the other two APIs. Then I will transform the data by standardizing artist names and album titles so they are consistent across each source. Then I will use the standardized data to match records between the APIs, using the artist names and album titles as primary keys. This will load all three APIs into one large schema which can then be processed for music discovery applications.